LinkSphere

Architecture & Hierarchy Blueprint

One place to save, organise, share and monitor links.
Modules, hierarchy, data, permissions and flowcharts, explained simply.

1	Overview and modules	6	Metrics
2	System architecture	7	Link health rules
3	Application hierarchy	8	Flowcharts (Diagrams 3 to 10)
4	Resources and permissions	9	How modules work together
5	Data model	10	Implementation sequence

How to read the diagrams



Boxes show steps, diamonds show yes/no decisions, and colours show who performs the step.

Version 1.0 · October 2026 · Technology-neutral design

1 Overview

LinkSphere is a central place to save and organise links. Personal use and team use sit side by side.

Who uses it

Role	What they do
User (student, teacher, employee)	Saves links in personal categories, joins teams, searches, creates short URLs.
Team Manager	A user who runs a team: members, roles, shared categories and folders. Company management uses this role for their teams.
Admin	Separate admin login. Manages user accounts, teams and platform resources; sees platform-wide metrics.

Design rules

- Two separate entrances: **User Login** and **Admin Login**.
- Every action passes a permission check first.
- Link health is shown **only** on dashboards and link cards. No alerts, warnings or notifications.
- One module has one job. Modules share information through the data store.
- Archived and deleted links are never health-checked.

Modules and their jobs

Module	Responsibility	Triggered by
Authentication & Access Control	Login, roles, permission check on every request	User / Admin
Link Module	Add, edit, open, copy, archive, restore, delete links; favorites	User
Category & Folder Module	Create, rename, delete categories and folders; move links	User
Team Module	Teams, members, roles, shared categories and folders	User
Search Module	Find links by name, category and tags	User
Short URL Module	Create a unique code, store the mapping, redirect visitors	User + public
Link Health Checker	Checks active links every 5 minutes and when a link is added; classifies results	Automatic
Metrics Module	Calculates dashboard counts; refreshed after every relevant change	Automatic
Admin Module	User accounts, teams, platform resources, platform metrics	Admin

2 System Architecture

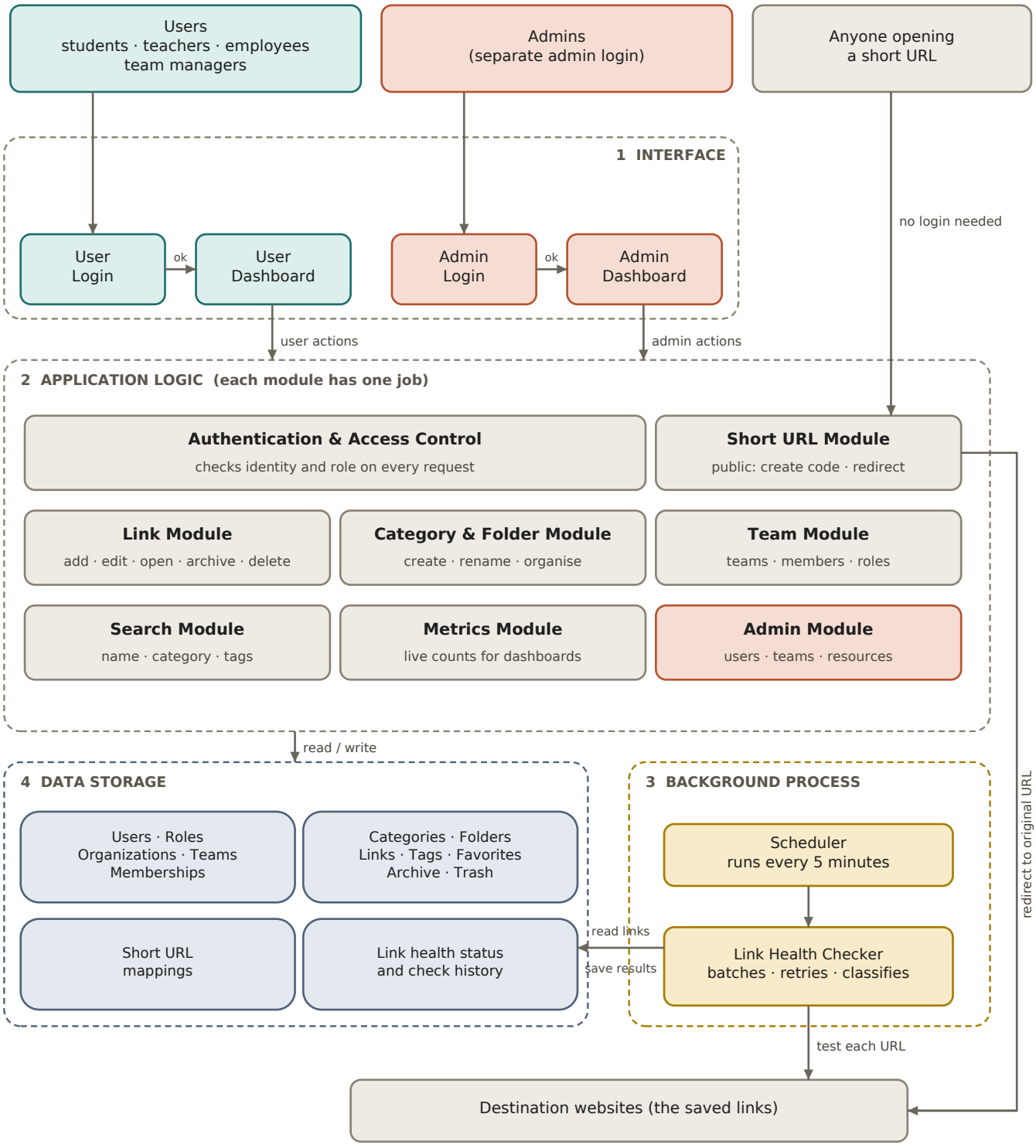


Diagram 1: High-level system architecture

- Read it top to bottom: **people, interface, modules, data**.
- The **Link Health Checker** runs on its own schedule and is the only part that contacts outside websites.
- The **Short URL redirect** is public. Every other request goes through Authentication & Access Control.

3 Application Hierarchy

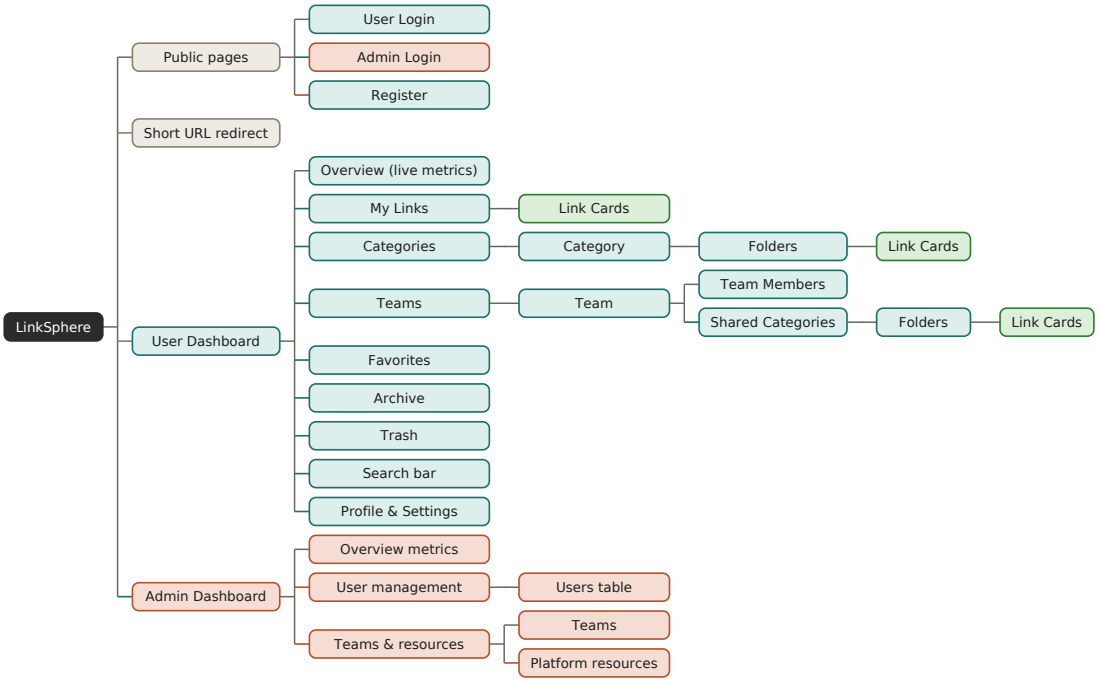
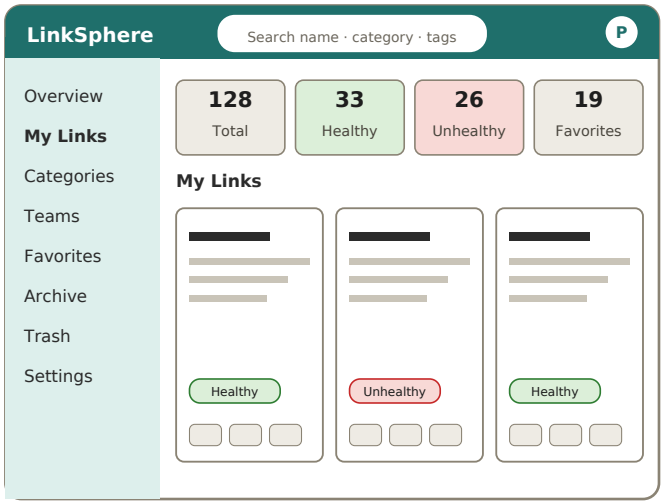


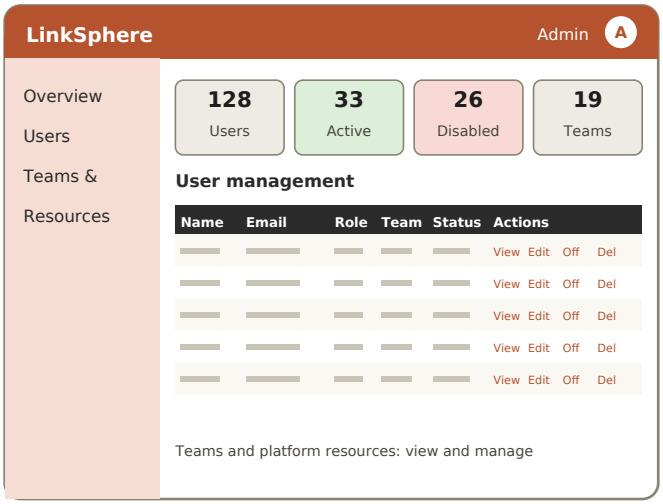
Diagram 2: Complete application hierarchy (pages, sections, categories, folders, teams, link cards)

Folders: create, rename, delete. **Team Members**: add, remove, set role. **Users table**: view, edit, disable, delete. **Search bar**: name, category, tags; visible on every user page.

3 Dashboard Layouts and Link Card



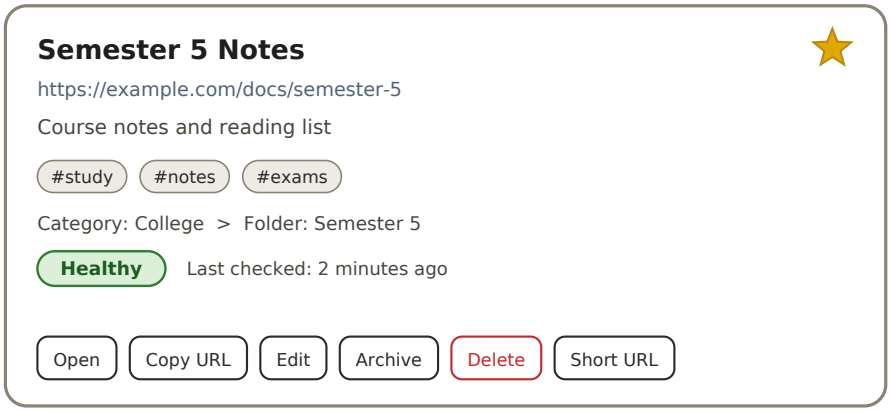
User dashboard: top bar (search, profile), side menu, live metric cards, link cards.



Admin dashboard: platform metrics, user table with row actions, teams and resources.

Diagram 2B: Dashboard hierarchies (wireframes)

Link card



Sample link card layout

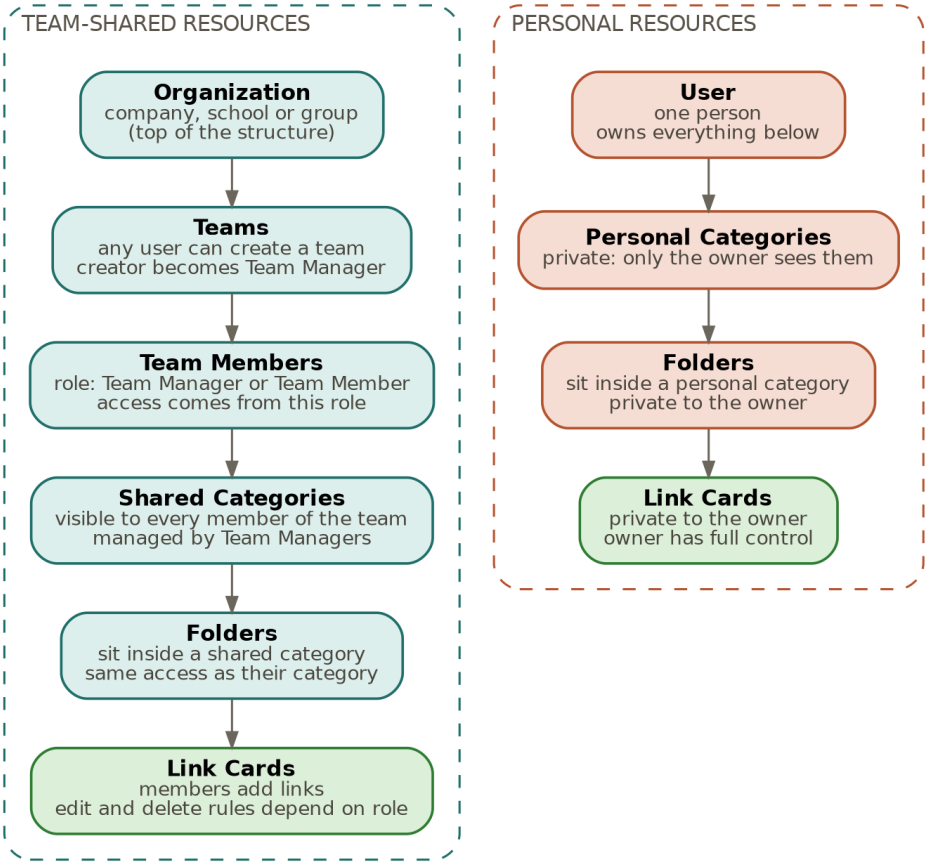
Every card shows
- Title and favorite star
- Original URL and description
- Tags, category and folder
- Health status and last checked time

Every card offers
- Open and Copy URL
- Edit, Archive, Delete
- Generate short URL, then copy it separately

Health badges

Badge	Meaning
Healthy (green)	The link responded successfully.
Unhealthy (red)	A failure was confirmed.
Restricted / Not verified (grey)	The site limits automated access, or the link has not been checked yet. Not counted as a failure.

4 Resources and Permissions



Resource hierarchy: team-shared (left) and personal (right)

Personal vs team-shared

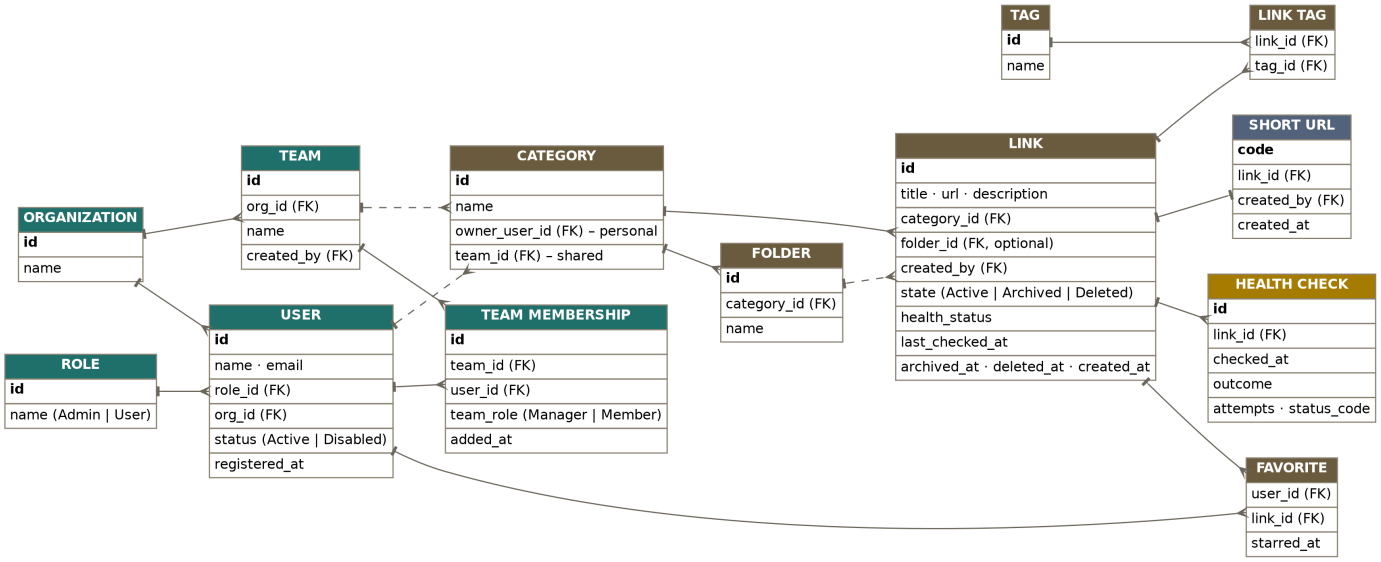
	Personal resources	Team-shared resources
Owned by	One user	A team inside an organization
Who can see	Only the owner	Every member of the team
Who can change	Owner: full control	Managers: everything. Members: add links, change their own links
Where it appears	My Links, Categories	Teams, then Shared Categories

Permissions

Action	Member	Manager	Admin
View shared categories, folders, links	Yes	Yes	Yes
Add links to shared folders	Yes	Yes	Yes
Edit, archive, delete own links	Yes	Yes	Yes
Edit, archive, delete other people's links	No	Yes	Yes
Create, rename, delete shared categories and folders	No	Yes	Yes
Add or remove members, change roles, delete team	No	Yes	Yes
Open admin dashboard, manage user accounts	No	No	Yes

Any user can create a team and becomes its Manager. Assumption: admins see counts for personal links, not their contents.

5 Data Model



Entity relationship diagram (solid = required link, dashed = optional / either-or, crow's foot = many)

- **Link states**: Active, Archived, Deleted. Archive and Trash are the same table with a different state and timestamp.
- **Restore** sets the state back to Active.
- **Category owner**: either one user (personal) or one team (shared).
- **Favorites** belong to each user, so one person's star never affects another.
- **Health**: the latest result is kept on the link; every check is also kept as history.
- **Short URL**: one code per link.

6 Metrics

User dashboard

Metric	What it counts
Total links	Active links the user can see (personal and team)
Healthy / Unhealthy	Active links by their latest health result
Not verified	Links not checked yet, plus Restricted links
Favorites	Starred active links
Archived / In Trash	Links in the Archive and in the Trash
Categories / Teams	Categories owned and teams joined

Admin dashboard

Metric	What it counts
Total registered users	All accounts
Active / Disabled users	Accounts by status
Total teams	All teams
Total links	All active links on the platform
Healthy / Unhealthy links	Active links by latest health result
Platform activity	New users this week, links added today and this week, links checked in the last cycle

How metrics change

Event	Total	Healthy / Unhealthy / Not verified	Archived	Trash
Link added	+1	Starts as Not verified; instant check moves it to Healthy or Unhealthy		
Link rechecked		Moves between Healthy, Unhealthy, Not verified if the result changed		
Link archived	-1	-1 from its current group	+1	
Link deleted (to Trash)	-1	-1 from its current group		+1
Link restored	+1	Not verified, then instant check sets the group	-1 if from Archive	-1 if from Trash
Star added or removed		Only Favorites changes (+1 / -1)		

All counts are calculated from stored data after each change, so a dashboard never shows stale numbers.

7 Link Health Rules

Check result	Meaning	Shown as	Counted as
2xx, or redirect that ends in 2xx	Link works	**Healthy** (green)	Healthy
404, 410, address not found, refused, or server error on every try	Confirmed failure	**Unhealthy** (red)	Unhealthy
Invalid web address	Cannot work	**Unhealthy** (red)	Unhealthy
401, 403, 429	Site blocks automated checks; link may still work for people	**Restricted** (grey)	Not verified
Timeout, mixed errors, network trouble	Temporary or unclear	Last status kept	Last status

Rules

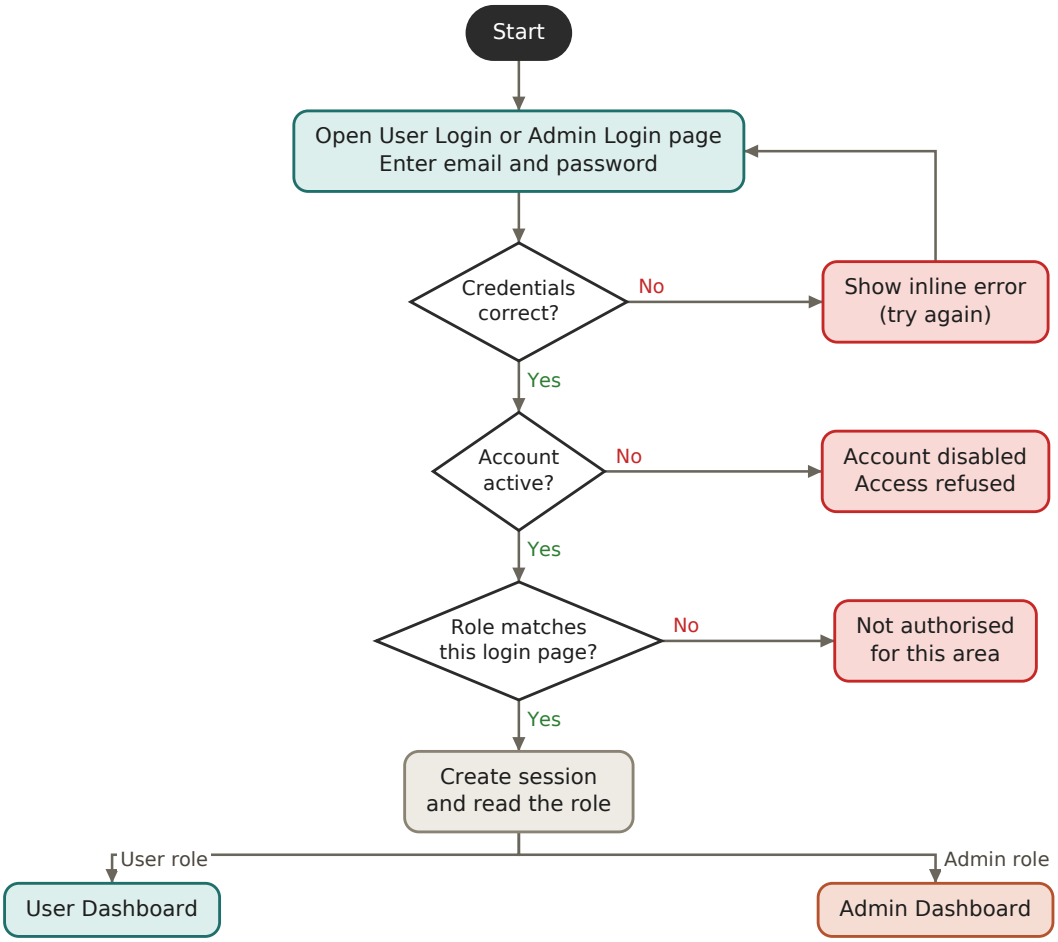
- **When**: every 5 minutes for all Active links, and immediately for a link that is added, edited or restored.
- **Retries**: up to 3 tries with a short wait. Failure is confirmed only if the same definite failure repeats.
- **Uncertain 3 times in a row** (about 15 minutes): mark Unhealthy, so a dead link never looks healthy forever.
- **No overlap**: if the previous run is still going, the next tick is skipped.
- **Batches**: links are checked in small fixed-size groups, so large collections stay manageable.
- **Excluded**: Archived and Deleted links.
- **Output**: status and last-checked time are saved; each check is added to history; metrics are recalculated. No notifications.

Why Restricted is not Unhealthy

Many sites answer automated requests with 'forbidden' while working fine in a browser. Calling those links broken would flood users with false failures.

8 Flowcharts

Diagram 3: User authentication



Login page	Accepts role	Lands on
User Login	User (including Team Managers)	User Dashboard
Admin Login	Admin only	Admin Dashboard

- A wrong email or a wrong password gives the same inline error, so the form never reveals which one was wrong.
- A disabled account cannot sign in, and its open sessions end.
- Every later request re-checks the role, so a normal user cannot reach admin pages by typing an address.

Diagram 4: Link CRUD (add, validate, save, edit, open, copy, archive, restore, delete)

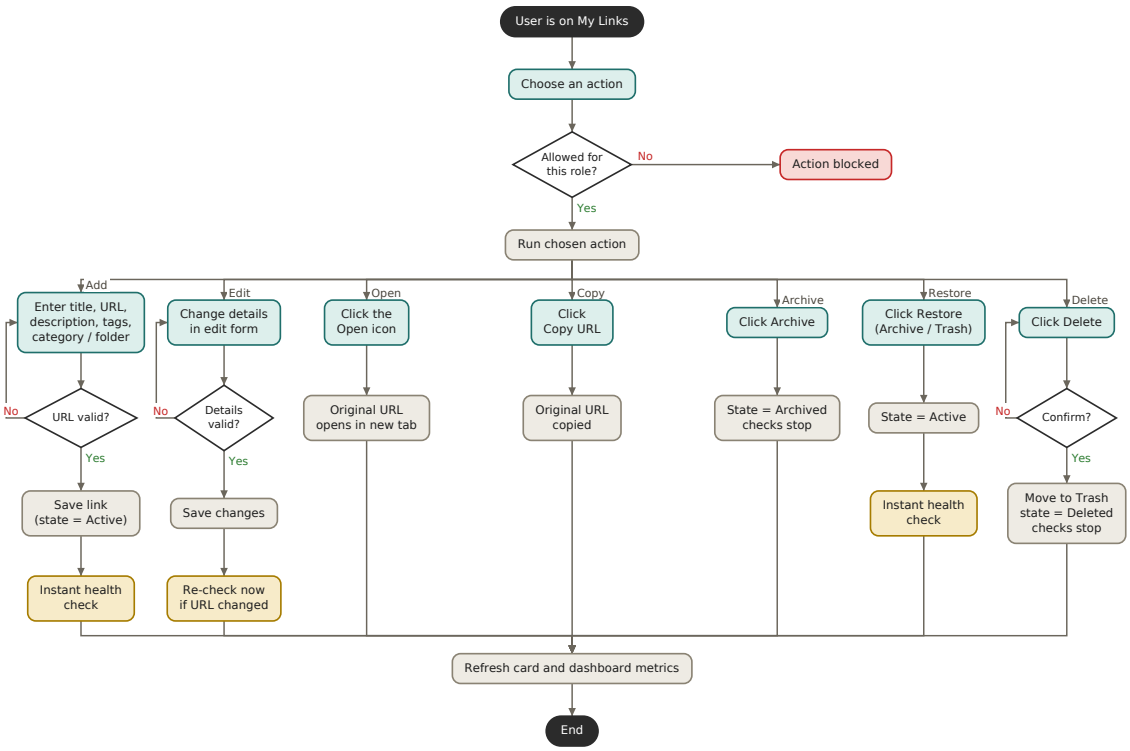


Diagram 5: Categories and folders

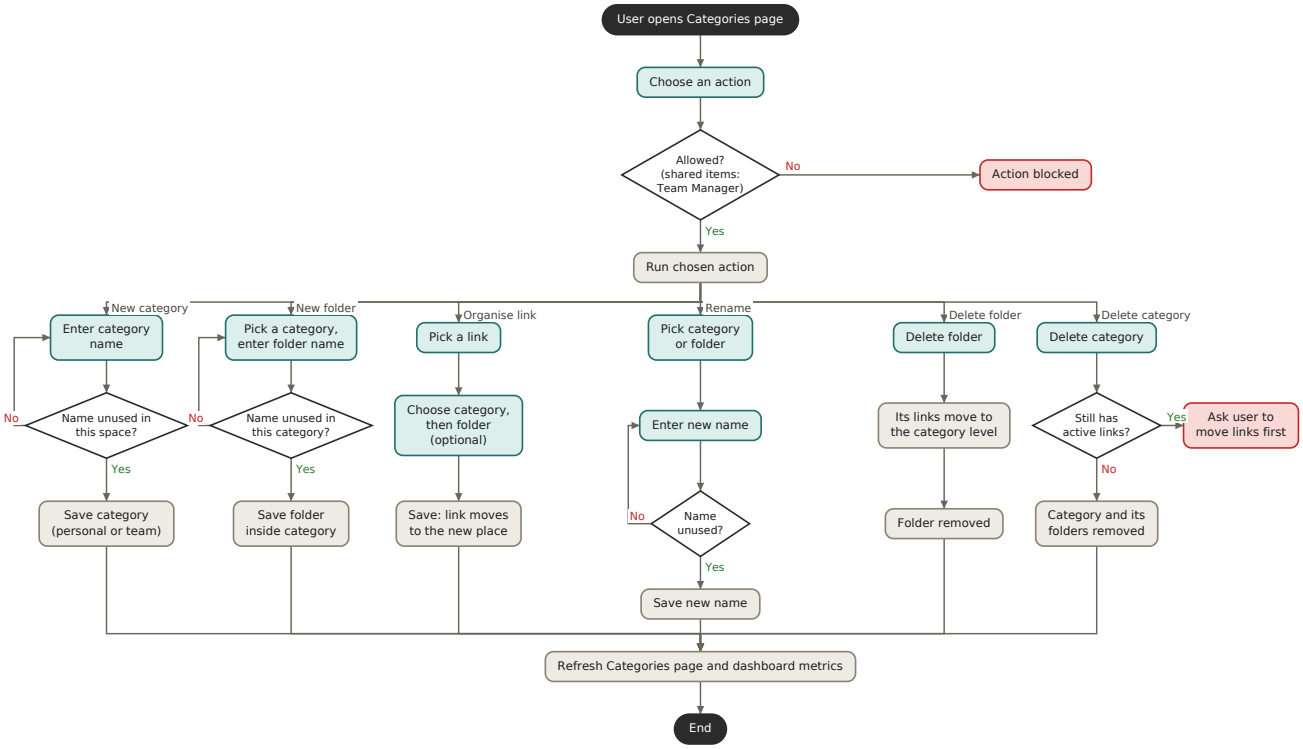


Diagram 6: Team management

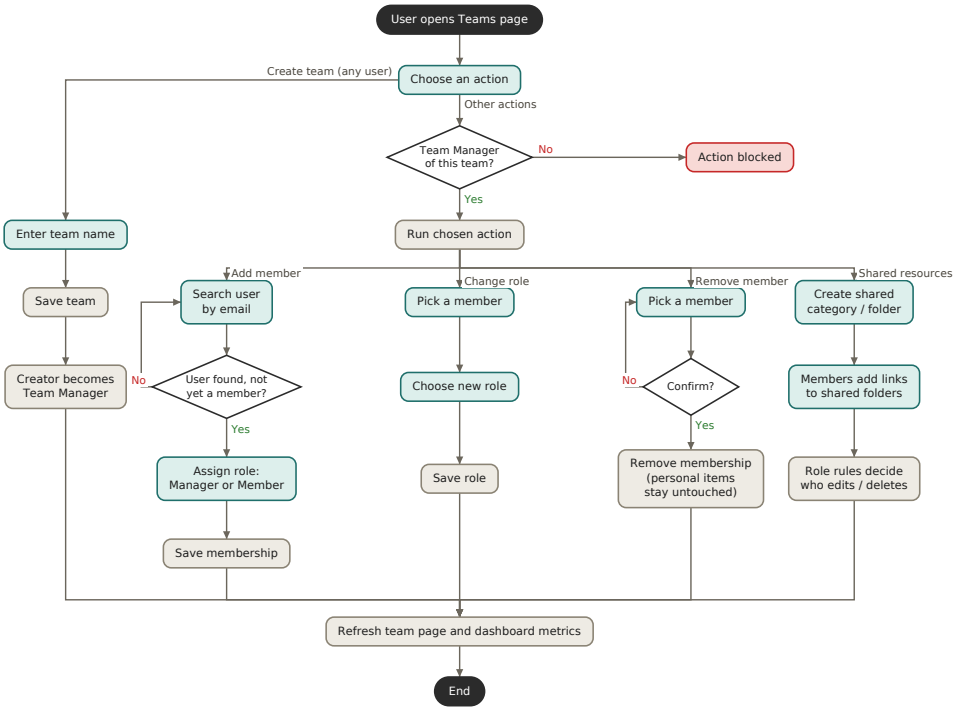


Diagram 7: Automatic health check

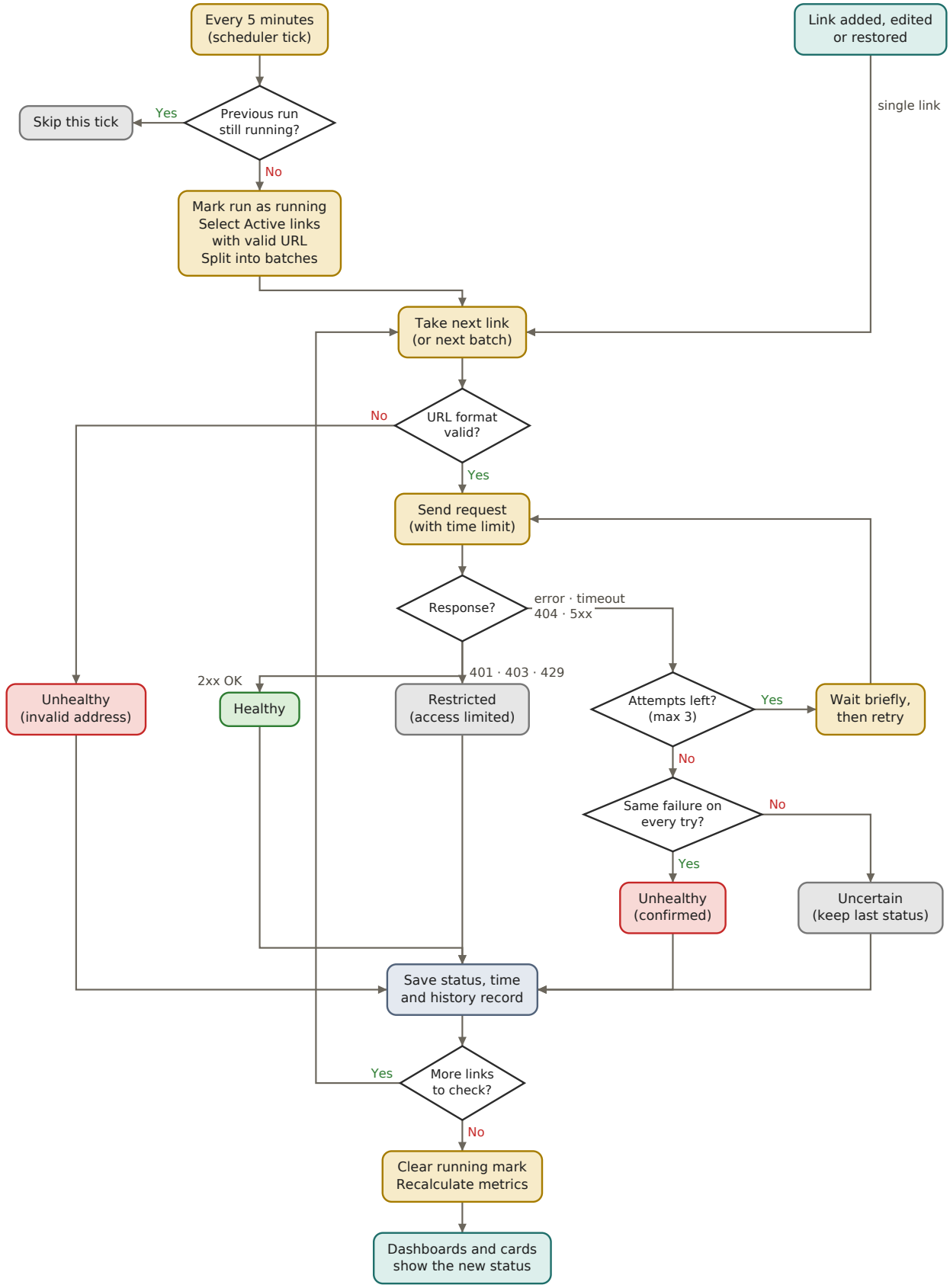
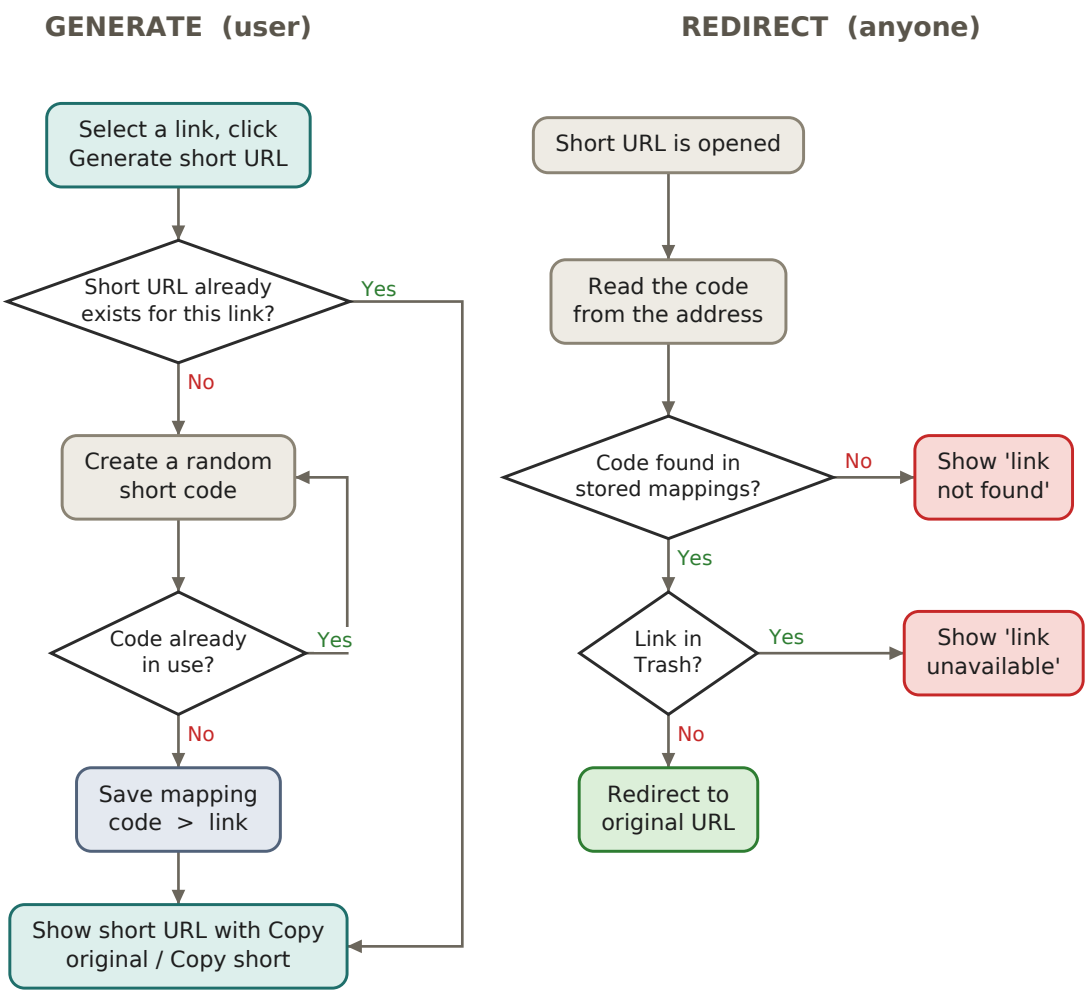
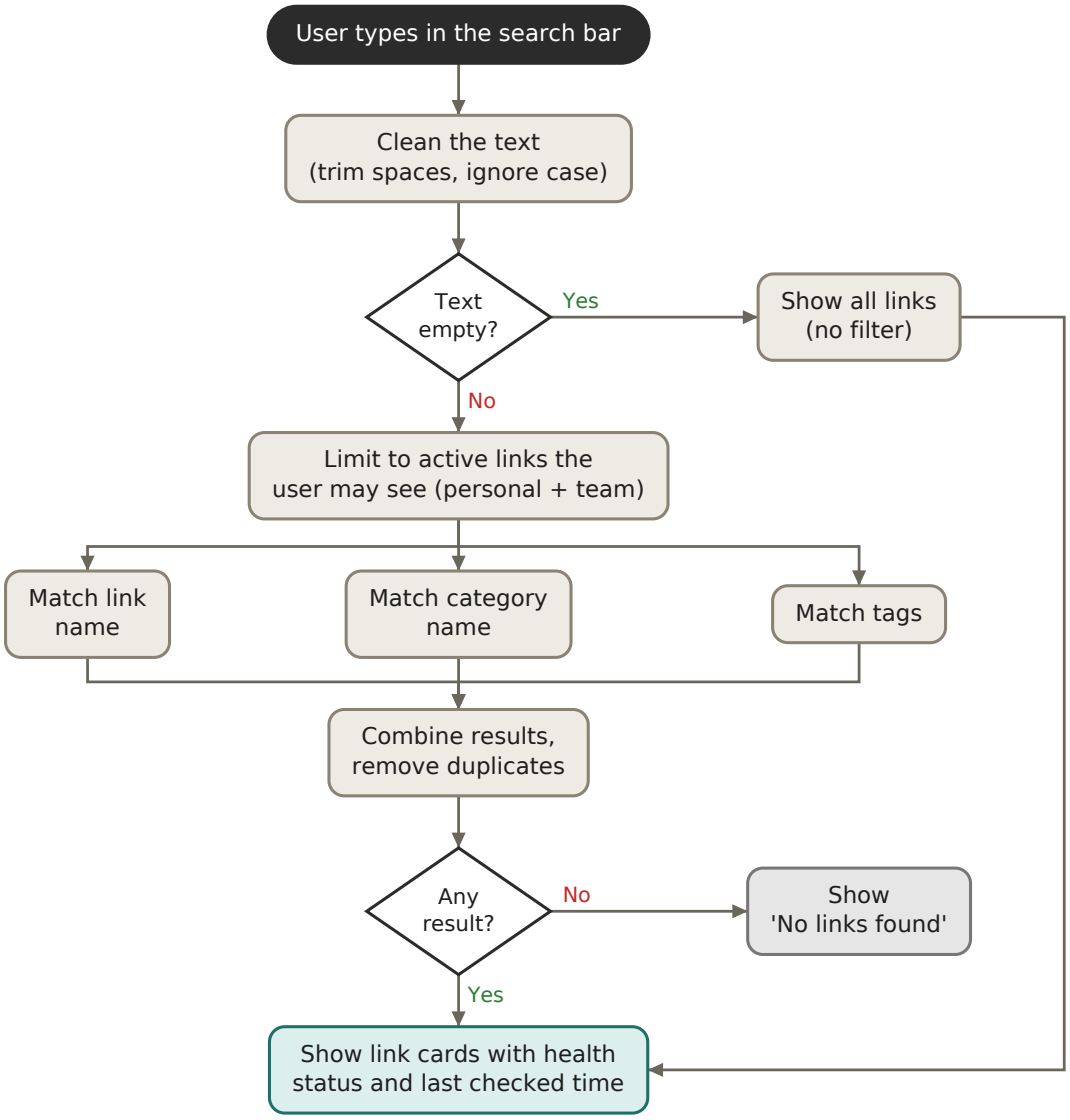


Diagram 8: Short URL generation and redirect



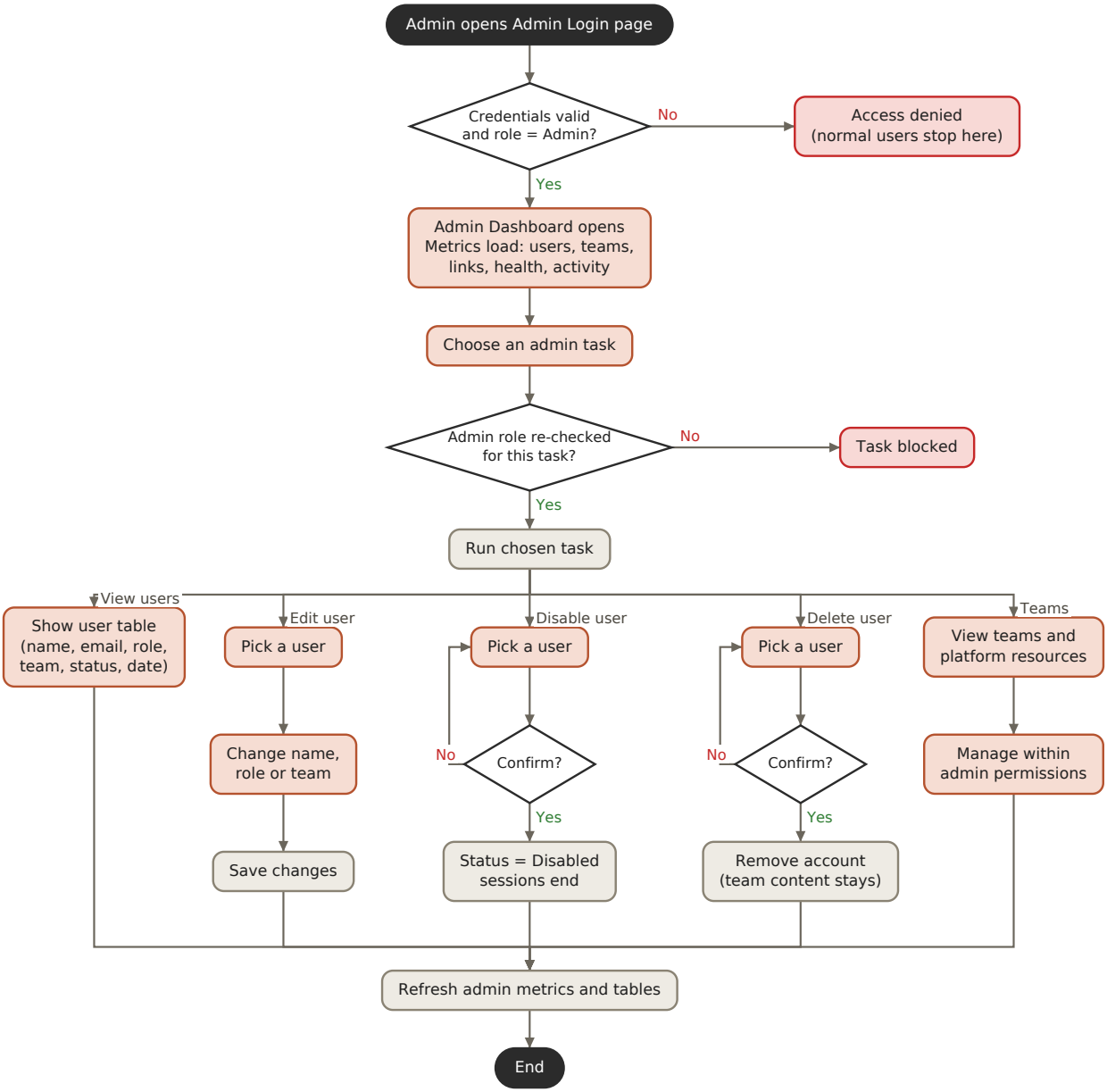
- A short code is random and checked against stored codes; if taken, a new one is made.
- The mapping (code to link) is saved once. Both the original and the short URL have their own Copy button.
- A link in Trash cannot be reached through its short URL. Restoring the link makes it work again.

Diagram 9: Search



- Search covers active links the user may see: personal links and links in the user's teams.
- Results show the same link cards, including health status and last checked time.

Diagram 10: Admin management



9 How Modules Work Together

Add a link



Automatic health cycle



Open a short URL



Admin disables a user



10 Implementation Sequence

Step	Build	Why this order
1	Data model, registration, login, roles, permission checks	Everything depends on knowing who the user is
2	Categories, folders, link cards with add, edit, open, copy	The core value: saving and organising links
3	Tags, favorites, archive, trash and restore	Completes the link life cycle
4	Search by name, category, tags	Useful once there are links to find
5	Dashboard overview metrics	Reads data that already exists
6	Health checker: instant check, then the 5-minute schedule, retries, classification, history	Needs stored links; adds Healthy / Unhealthy to cards and metrics
7	Teams, memberships, shared categories and folders, role rules	Extends personal rules to groups
8	Short URL generation and redirect	Independent feature; needs stable links
9	Admin dashboard, user management, platform metrics	Needs every other entity to report on
10	Review permissions, edge cases, large-collection checks	Final safety pass

Design decisions to confirm

- Company management uses the Team Manager role; platform-wide oversight is the Admin role.
- Admins see counts for personal links, not their contents.
- A restored link is health-checked immediately.
- A category with active links cannot be deleted; deleting a folder moves its links up to the category.
- Three uncertain checks in a row turn a link Unhealthy.